

Text2Groove: Text-Guided Drum MIDI Humanization

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Abstract—Programmed drum performances often require extensive manual adjustment of timing and dynamics to achieve a desired groove. This paper presents Text2Groove, a text-guided MIDI humanization system that generates expressive timing offsets and velocities for a given quantized drum pattern and natural language prompt. A Transformer-based Conditional Variational Autoencoder is trained on the Groove MIDI Dataset, augmented using nearest-neighbour groove transfer, and conditioned on genre, feel, and dynamics. Objective results show accurate reconstruction of human performance parameters and effective control of dynamics, while control of rhythmic feel, particularly swing, remains more difficult. Listening tests found that when compared to the groove template baseline, Text2Groove achieved very similar observed prompt-matching ratings but was rated worse on human-likeness. These results demonstrate the potential of natural-language control for drum MIDI humanization while highlighting limitations in semantic conditioning and synthesized training data.

Index Terms—Expressive performance modeling, Microtiming, MIDI humanization, Drum groove generation, Text-conditioned generation, Conditional Variational Encoder

I. INTRODUCTION

The same drum score can be performed in many different ways. By, varying elements such as microtiming and dynamics, a skilled drummer can leverage this flexibility to give a drum pattern different distinct grooves. For example, a pattern might be performed in a Funk style with a loose pocket and relatively soft dynamics, or the same score could be played in a Rock style with swung timing and much stronger dynamics. Both performances are valid interpretations of the same underlying score, yet they can convey entirely different musical characteristics. Achieving this level of variation is considerably more difficult for music producers working with programmed drums. Producers must manually adjust note timing, velocity, and other performance parameters to create a desired groove, a process that can be time-consuming and requires an understanding of how these characteristics interact. Even experienced producers have to “devote a lot of time and attention to the careful temporal positioning of sounds” (Brøvig-Hanssen et al. 2022) to achieve the exact groove they are looking for. With Text2Groove, we try to address this production challenge by training a machine learning model that takes in a drum score and a natural language prompt,

then generating offsets and velocities of each hit to convey the groove described in the given prompt.

II. BACKGROUND

A. Microtiming, Dynamics, and Tempo

Microtiming refers to small deviations of when notes are played, ahead or behind a strict temporal grid. These deviations occur at the millisecond level and do not change the musical score of the piece but make a difference in how the performance feels. For example, ahead of the beat may create urgency, while playing behind it can create a more relaxed feel. However, microtiming does not necessarily improve groove. Previous studies have found that exaggerated or systematic deviations can reduce perceived groove quality (Frühauf, Kopiez, and Platz 2013; Davies et al. 2013) and the effect of timing deviations depends on musical style (Senn et al. 2016). Real performances therefore involve more nuanced timing relationships than simply introducing random offsets or manually offsets.

Dynamics provide another important aspect of expressive performance. Rather than when the note is played, dynamics is concerned with how the note is played. Drummers use variations in velocity, accents, and ghost notes to shape the character of a groove and these patterns can differ across genres (Danielsen, Johansson, et al. 2023). Moreover, dynamics also interact with timing and rhythmic structure. Laid-back strokes may be played louder than on-beat strokes (Danielsen, Waadeland, et al. 2015), while more rhythmically complex patterns can result in stronger dynamics (Câmara et al. 2020). Finally, tempo can further influence both dynamics and the perception of swing (Danielsen, Waadeland, et al. 2015; Ainsworth 2025). The drum score, tempo, microtiming, and dynamics therefore interact continuously, with each element influencing the others to create a cohesive and expressive drum performance.

B. Existing Humanization Methods

Traditional MIDI humanization methods can broadly be divided into rule-based timing controls, groove transfer, and more recently, deep learning approaches. Rule-based methods such as the swing amount control implemented in Roger Linn’s LinnDrum and MPC designs (Scarth and Linn 2013) provide

simple timing control but fails to capture the variation found in human performances. Groove templates extract timing and velocity information from an existing performance and transfer it to another pattern (Cholakis 1999). While this can preserve realistic performance characteristics, the user must first find a suitable reference groove and has limited control over how the characteristics are applied.

More recent approaches use deep learning to model expressive performance directly from human data. GrooVAE (Gillick et al. 2019), for example, generates realistic drum microtiming and velocity from a drum score and was shown to perform comparably to human data in listening tests. However, its expressive output cannot be explicitly controlled using high-level descriptions such as genre, timing feel, or dynamics. Text2Groove extends this direction by conditioning expressive drum generation on natural-language descriptions of the desired performance.

C. Semantic Audio Production

Semantic audio production uses high-level descriptors to control audio processing rather than requiring users to manipulate low-level technical parameters. Previous systems have applied natural-language or semantic control to tasks such as equalization and audio effects (Seetharaman and Pardo 2016; Zheng, Seetharaman, and Pardo 2016; Chu et al. 2025). This can reduce the technical knowledge required to achieve a desired sound and allow users to describe perceptual goals directly. However, little research has been done on leveraging natural language descriptions to control MIDI performance parameters such as microtiming and velocity. Producers must typically adjust these parameters manually to achieve a desired rhythmic feel, requiring both technical knowledge and an understanding of how these parameters interact. Text2Groove therefore applies the principles of semantic audio production to expressive drum performance, translating natural-language descriptions into changes in timing and dynamics.

III. METHODOLOGY

A. Design Overview

Text2Groove takes a quantized drum pattern, natural language prompt, and tempo as input and generates timing offsets and velocities while leaving the original hit pattern unchanged. The conditioning space is limited to four genres (Funk, Rock, Hip Hop and Jazz), three feels (tight, loose and swung), and three dynamic levels (soft, medium and loud) to evaluate the feasibility of semantically controlled drum humanization. Training data is derived from the Groove MIDI Dataset and augmented using nearest neighbour groove transfer so that similar rhythmic material can appear with different expressive characteristics. Then, a Transformer-based Conditional Variational Autoencoder (CVAE) is used to model the many possible performances that can correspond to the same score and prompt, with the latent variable z capturing variation not explicitly described by the conditioning. The model is evaluated on reconstruction accuracy, prompt controllability, latent diversity, and perceptual quality through a listening test.

B. Implementation

1) *Dataset*: The dataset used to train the model is the Groove MIDI Dataset by Google Magenta introduced in (Gillick et al. 2019). It is composed of 13 hours of aligned MIDI and audio of human-performed expressive drumming performed by a total of ten drummers. Each performance is annotated with a genre provided by the drummer, tempo, time signature, whether the performance is a beat or a fill and which split (train/validation/test) it belongs to. This work focuses on beat performances in 4/4 time signature.

2) *Preprocessing*: Preprocessing follows the same method described in (Gillick et al. 2019). All drum hits are mapped to a smaller set of 9 canonical drum categories, following Roberts et al. (Roberts et al. 2018). To create the drum patterns, we slide fixed sized windows of 2 bars across all MIDI files with a hop size of 1 bar. This was chosen because 2 bar patterns are long enough to contain sufficient variation but short enough to quickly evaluate in listening tests. Each drum hit is associated with the closest 16th note metrical position. If multiple hits are in the same drum category and the same timestep, we keep the loudest one (the one with the highest velocity). These patterns are then represented with three 32x9 matrices, one for drum hits (H), one for offsets (O) and one for velocities (V). H is a binary matrix that stores the presence or absence of drum onsets. Each column contains the drum score for one of the nine drum voices throughout the two bars while each row contains the drum score for all nine instruments at a single timestep. O is a continuous-valued matrix that stores the timing offsets of each drum hit in H. The values range from $[-0.5, 0.5)$ and they indicate how far and in which direction each note lies relative to the nearest 16th note. Positive values indicate playing behind the beat (late) where negative values indicate playing ahead (early). V is another continuous-valued matrix that stores velocity information converted from MIDI domain (0 to 127) to real numbers in $[0, 1]$.

3) *Synthesized Data*: Early training attempts showed that the model generated outputs primarily from the input hit pattern while ignoring the accompanying prompt. This behaviour likely arose because most patterns appeared in the training data with only a single humanized performance. During data collection, the drummers were instructed to play freely, provided that they remained in time with the metronome. Consequently, multiple humanized versions of the same pattern occurred only by chance, such as when different drummers happened to perform an identical pattern with a different genre, timing, or feel. The model therefore had limited evidence from which to learn how a given pattern could be transformed in response to different prompts. Instead, it tended to reproduce the specific interpretation present in the training data rather than associate the prompt with alternative variations in timing and dynamics. To solve this issue, synthesized variants of each pattern was created to add to the training data. The different variants are combinations of the 4 genres, 3 feels and 3 dynamics. To ensure that these new variants still sounded realistic and played by real human drummers, they were created by finding another

pattern in the dataset with the most similar hit pattern, played in a similar tempo and fit the target criteria, then transferring the offsets and velocities to the original hit pattern. Since the Groove MIDI Dataset come annotated with the genre for each MIDI recording, we simply use the genre annotation to determine the genre of the pattern. To determine if a pattern fits the feel and dynamics criteria, additional groove features were measured.

Subdivision: First, we determine if a pattern is played primarily in 8th notes or 16th notes by counting the number of valid 8th note swing pairs and the number of 16th note swing pairs. This helps us calculate the mean swing ratio of the pattern at the correct swing level. More specifically, for 8th notes, we count the number of valid pairs with timesteps (0,2), (4,6), (8,10), ... These refer to the on beats and offbeats at the 8th note level. Similarly, for 16th notes, we count the number of valid pairs with timesteps (0,1), (2,3), (4,5), ... These refer to the on beats and offbeats at the 16th note level.

Mean Swing Ratio: With the subdivision, we can then calculate the mean swing ratio according to previous work (Frane 2017; Ainsworth 2025). Local swing ratios were calculated by comparing the Inter-Onset Interval (IOI) between an on beat note and the following off beat note at the subdivision level of the pattern. The swing ratio of the pattern was then computed as the mean of its local swing ratios. A triplet swing corresponds to a swing ratio of 2, where the duration of the on beat note is twice as long as the following off beat note. This swing ratio value was used to determine if a pattern satisfies the swing criteria. While there has been little research on the threshold that separates straight and swing (Frane and Shams 2017), Butterfield (2011) loosely estimated that the noticeable difference in swing ratio is somewhere between 1.1 and 1.2. For the sake of being able to easily perceive swung patterns from the model, a threshold of 1.5 was chosen to determine if a pattern is swung. The chosen value is not as intense as a triplet swing with a ratio of 2, but not as subtle as 1.1 or 1.2.

Standard Deviation of Offsets: The standard deviation of the offsets of each drum hit is calculated and used to estimate whether a pattern is played loose or tight. Based on the mean standard deviations of patterns categorized by genre in the Groove MIDI Dataset, an arbitrary threshold of 0.1 was used. If a pattern has a standard deviation of offsets over 0.1, it is considered loose, otherwise it is considered tight.

Mean Velocity: The mean velocity of the drum hits was calculated to estimate the overall dynamics of each pattern, classifying patterns as soft, medium, or loud. The classification thresholds were defined using the 33rd and 67th percentiles of the distribution of mean velocities across all patterns in the dataset. The exact value of the chosen values are 0.4832396 for soft, 0.6237068 for loud, and all mean velocities in the between those two values are defined as medium.

The final training dataset contains 144,696 samples, with each sample containing the pattern represented by the three matrices H , O and V , along with metadata of the pattern’s tempo, genre, feel and overall dynamics. The dataset is then split into a training set of 115,758 samples, a validation set of

14,469 samples and a test set of 14,469 samples. It is worth noting however, that while all other combinations of genre and feel has over 10,000 samples, Hip Hop swung has significantly less samples at only 942. This has to do with the amount of swung Hip Hop data in the original Groove MIDI Dataset, with Hip Hop having the least amount of performances out of the four chosen genres and even less swung Hip Hop performances. As a result, we expect swung Hip Hop outputs from the model to perform the worst due to the lack of training data.

4) *Text and Tempo Conditioning:* Combinations of genre, feel and dynamics labels were converted into natural language prompts using a few different wording variations and prompt patterns. For example, a pattern that is Funk, tight and soft could be associated with the prompt "Make this sound like Funk with tight, precise timing, play it softly." or "Give this a Funk groove locked to the grid, use gentle dynamics.". These variations are randomly assigned for each groove with the same genre, feel and dynamics to prevent the model from relying on one fixed prompt format. The text is then encoded using the pretrained all-MiniLM-L6-v2 Sentence Transformer, producing a 384 dimension representation, which is then projected to 192 dimensions. The tempo was normalized and also projected to 192 dimensions, then it was added to the text embedding to form the conditioning vector C .

5) *Conditional Variational Autoencoder:* The proposed model is a CVAE containing a posterior encoder and a performance decoder. Figure 1 shows a high-level diagram of its architecture. During training, the posterior encoder approximates

$$q_{\phi}(z | H, O, V, C) \quad (1)$$

where H, O, V are the hit, offset, and velocity matrices respectively, and C is the combined text and tempo conditioning. The three matrices H, O, V are stacked into a three dimensional feature vector, resulting in a sequence of 288 feature vectors, one for each matrix. Each feature vector is then independently projected to a 192-dimensional representation, forming the 288 input tokens processed by the Transformer. Each token is then supplemented with three learned embeddings representing its timestep, drum voice, and metrical position within a bar. Since each pattern is 2 bars of 16th notes, the metrical position goes from 0 to 15 for the first bar then repeating from 0 to 15 for the second bar. This allows the model to learn that corresponding positions across the two bars share the same metrical role. Instead of representing the combined text and tempo conditioning vector as a single conditioning token, it is instead added to every token before being passed to the posterior encoder. This encourages the network to account for the requested performance characteristics throughout the entire pattern, reducing the likelihood of the model ignoring the conditioning prompt, as observed in earlier iterations. The conditioned sequence is then preceded by a learned posterior token, similar to the [CLS] token used in BERT (Devlin et al. 2019), and processed by a three-layer Transformer encoder with a model dimension of 192, six attention heads, a feed-

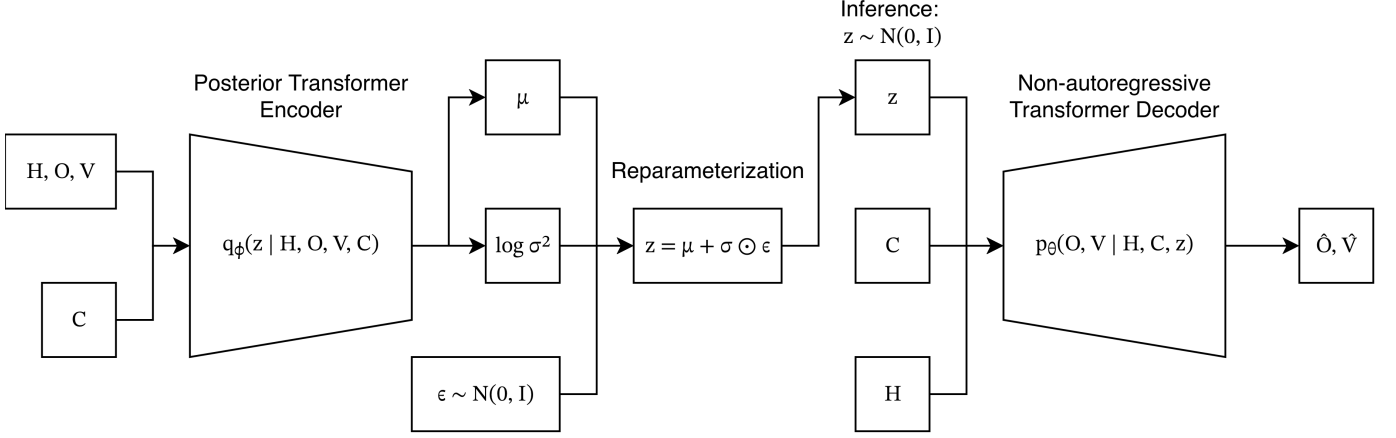


Fig. 1. Diagram of the Text2Groove CVAE architecture.

forward dimension of 768, and dropout of 0.1. The resulting posterior token acts as a summary of the input performance and is passed through two linear layers to produce the 16-dimensional latent mean μ and log variance $\log \sigma^2$. A latent representation is then sampled using the reparameterization trick described by Kingma and Welling (2014),

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \quad (2)$$

which allows gradients to propagate through the encoder during training.

On the other hand, the decoder models

$$p_{\theta}(O, V | H, C, z) \quad (3)$$

It predicts the offsets and velocities based on the given hit pattern H , text and tempo conditioning C and the sampled latent representation z . The hit value matrix goes through a similar process as in the encoder. Each hit is projected to the 192-dimensional model space and combined with the same timestep, voice and metrical embeddings used by the encoder. The conditioning vector C and latent representation z are then incorporated into each token, providing the decoder with the requested performance characteristics and which particular variation of it to generate. The resulting sequence is processed by another three-layer Transformer with the same dimensionality and attention configuration as the posterior encoder. The decoder is non-autoregressive, meaning that all offsets and velocities are predicted simultaneously rather than predicted one timestep at a time. This is suitable for this humanization task since the hit pattern is known in advance and is not modified by the model at all. Finally, two output heads predict the timing offsets and velocities. A hyperbolic tangent activation constrains the predicted offsets to $[-0.5, 0.5]$ while a sigmoid activation constrains velocity predictions to $[0, 1]$, matching how the training data is represented. At inference time, z is sampled directly from the unit Gaussian prior, $z \sim \mathcal{N}(0, I)$, and is passed to the decoder along with the hit pattern, prompt and tempo. Therefore, sampling

different values of z will produce multiple possible expressive performances for the same input score and description.

6) *Training*: The model was trained using a combination of reconstruction loss and Kullback-Leibler (KL) divergence. Since offsets and velocities are meaningful only at positions with drum hits, reconstruction losses were masked using the hit matrix H . Huber loss was chosen because it behaves similarly to squared error for small prediction errors while being less sensitive to larger errors than mean squared error. The final reconstruction loss is the sum of the Huber loss for offsets and Huber loss for velocities, with equal weighting.

$$\mathcal{L}_{recon} = \frac{\sum_i H_i \text{Huber}(\hat{O}_i, O_i)}{\sum_i H_i} + \frac{\sum_i H_i \text{Huber}(\hat{V}_i, V_i)}{\sum_i H_i} \quad (4)$$

The latent distribution was regularized towards a unit Gaussian prior using the KL divergence,

$$\mathcal{L}_{KL} = -\frac{1}{2} \mathbb{E} [1 + \log \sigma^2 - \mu^2 - \sigma^2]. \quad (5)$$

The complete training objective was

$$\mathcal{L} = \mathcal{L}_{recon} + \beta \mathcal{L}_{KL} \quad (6)$$

where the maximum KL weight was set to $\beta = 0.01$. To reduce the risk of posterior collapse, β was linearly increased from 0 to 0.01 over the first 15 epochs. Training used the AdamW optimizer with a learning rate of 2×10^{-4} , weight decay of 10^{-4} , a batch size of 64, and gradient clipping with a maximum norm of 1.0. The pretrained all-MiniLM-L6-v2 Sentence Transformer remained frozen during training. Furthermore, genre-based weighted sampling was used to reduce the effect of class imbalance in the training data. To speed up the training process, validation was performed every five epochs, with a total of 50, and the checkpoint with the lowest validation objective after KL warm-up was saved.

C. Evaluation

To evaluate the model, quantitative metrics were used to evaluate the model's ability to reconstruct the original data, whether or not the model was able to interpret the given

prompts correctly in terms of feel and overall dynamics, and since the model is a VAE, we also quantify whether the latent variable actually creates diversity. Genre control was not tested using objective evaluation methods due to the lack of scalar features that quantifies a genre. It is instead tested through subjective listening tests. The listening tests were conducted online and evaluate how well the generated outputs match the given prompt and how likely they are to be played by a human drummer compared to other methods. We recruited 12 participants for the study through social media and forums relating to drumming and music production. Musical experience was evenly distributed between 1-5, 6-10 and 11 or more years of active practice, with four participants in each category.

1) Objective Evaluation: These tests were performed on the entire test set, with the exception of the latent diversity test, which randomly samples 1500 test patterns, a bit over 10 percent of the entire test set.

Reconstruction: For reconstruction, we test the offset and velocity masked Huber loss, which is used for training, mean absolute error (MAE), and Pearson correlation between the predicted outputs and the actual data from the test set. The MAE is also reported in milliseconds and MIDI units for offsets and velocities respectively.

Dynamics: We test dynamics control by calculating the mean velocities of outputs from prompts requesting soft, medium and loud, as well as validating the accuracy of each dynamic level using the previously determined mean velocity thresholds used to label the dataset. For each hit pattern, the tempo, genre, feel, and z stays constant, we generate three versions with the three different dynamic levels and evaluate as described above.

Feel: Same as above but for feel instead of dynamics.

Latent Diversity: Finally, we test the model’s latent diversity by randomly sampling 1500 test samples, then for each sample, fixing the hit pattern, the tempo, genre, feel and dynamic and generating 20 different versions with varying z . The predicted offsets and velocities should change if z actually contributes to generation. We then calculate and report the mean offset diversity and mean velocity diversity. This is done by taking the standard deviations of each drum hit across all 20 versions, then taking the average of those standard deviations. We also compare each unique pair in the 20 versions to find their mean differences in offsets and velocities, then computing the mean of those values across the 20 versions.

2) Listening Test: The goal of the listening test was to evaluate how well the model’s output matches the given prompt and how likely the model’s output was played by a human drummer. The proposed model was compared against three other humanization methods, the quantized score with constant velocity, the groove transfer method used to create the synthesized dataset, and groove templates found in Ableton and Reason. On each trial, the participants were presented with a target description specifying the genre, feel and dynamics, and four anonymous audio renderings of the same hit pattern humanized according to the same target description, but using

the four different methods. These descriptions are taken from the prompts in the training data. The participant then proceeds to rate how well each audio matches the target description and how likely each audio are likely to be played by a human. Both are done on a continuous scale ranging from 0, meaning not at all, to 100, meaning extremely, as shown in Figure 2. Similar to the study by De Man and Reiss (2017), participants were asked to drag markers representing each audio horizontally along a scale to rate them. Each participant completed 12 trials, comprising four distinct drum hit patterns and three target descriptions per pattern. Trials based on the same hit pattern were presented consecutively. The hit patterns and prompts each participant was presented with were completely randomized, as is the order of which the audios appear on the scale initially to remove any visual bias.

20 patterns were randomly selected from the test set and used for the listening test. For each of the 20 patterns and every training prompt, we create four versions of it using the four different methods. For the proposed method, we simply give the model the pattern and the prompt to get the offsets and velocities, then reconstruct the hits, offsets and velocities as a MIDI file, resulting in 19440 outputs. For the quantized method, each version are the same and has no microtiming changes with constant velocity, resulting in only 20 outputs. This is used as a lower anchor and is expected to perform the worst. The baseline version is created by applying groove templates found in Ableton and Reason to the 20 patterns. Ableton and Reason were chosen because they feature genre-labeled groove templates that are meant to capture certain characteristics of a genre, which we could then compare genre control with our model. For feel and dynamics, the same metrics were calculated for the groove templates and were labeled accordingly. Then for each combination of genre, feel and dynamic, we find one groove template that satisfies its criteria and apply it to the 20 hit patterns to generate that particular variant of the pattern. Since this method does not have to account for wording variations and prompt pattern variations, the final number of outputs for this method is 720. Finally, we use the same groove transfer method used to generate the additional training data for the fourth version. This is expected to receive the highest score. However, since the listening test requires each method to have an output for all combinations of genre, feel and dynamics for each pattern, the tempo and similarity constraints were removed. When generating the additional data, if the tempo or hit pattern similarity of the nearest neighbour pattern was too far off from the target pattern, meaning there exist no other performance in the dataset that played a similar hit pattern in this particular genre, feel and dynamic, at a similar tempo, it was excluded from the training data to prevent the model from seeing unrealistic results. However, they were included in this listening test so that each trial would always have four methods. Therefore, the quality of this method might be reduced. All generated MIDI files were rendered with Addictive Drums 2 by XLN Audio (2026) using a Reaper script (Cockos Incorporated 2026).

Target description

Make this sound like funk with swing, keep the intensity low.

Click each marker to listen. Drag every marker to your chosen position on both scales.
Each marker must be moved at least once before proceeding to the next trial.

How well does each audio match the target description?

Click a marker to listen. Drag it horizontally to rate.

Not at all
Slightly
Moderately
Very
Extremely

A B C D

How likely is each audio to have been played by a human drummer?

Click a marker to listen. Drag it horizontally to rate.

Not at all
Slightly
Moderately
Very
Extremely

A B C D

Fig. 2. An example trial of the listening test.

Prompt matching and human-likeness ratings were analyzed separately using linear mixed-effects models using the Python module statsmodels. Method, genre, feel, dynamics, trial order and marker position were entered as fixed effects with participant and drum pattern as random effects. Planned contrasts compared the proposed method with the baseline, nearest-neighbour, and quantized methods. For each of those comparisons, we compute and report the model-adjusted mean differences, 95% confidence intervals and two-sided p -values. Holm’s correction was applied to the three method comparisons within each outcome. Furthermore, we also investigated method by genre, method by feel and method by dynamics interactions. Each interaction model was compared with the main-effects model using a likelihood ratio test and the resulting three interaction p -values were Holm-adjusted within each outcome.

IV. RESULTS

Initial objective evaluation results showed that while the model was able to correctly predict the requested the dynamic about 70 to 80% of the time, it struggled with feel, particularly loose and swing with only 66.42% and 43.47% accuracy respectively. After some investigation, it was discovered that there was a bug when synthesizing the training data and some patterns were being classified as swing even if they are not actually swung. This led to a second iteration of the model with the bug fixed, along with a few other changes. After the bug fix, there are significantly less swung Funk and Hip Hop examples, which led us to redefine the swing threshold to 1.2 for those two genres, while keeping it 1.5 for Rock and Jazz. This was done in attempt to increase the amount of

	Huber Loss	MAE	MAE (ms/units)	correlation
T2G-1 O	0.0020	0.0370	5.2692 ms	0.9306
T2G-1 V	0.0027	0.0383	4.8731 units	0.9675
T2G-2 O	0.0014	0.0299	4.2827 ms	0.9401
T2G-2 V	0.0019	0.0299	3.8033 units	0.9766

TABLE I
RECONSTRUCTION METRICS OF OFFSETS O AND VELOCITIES V FOR BOTH MODELS.

swung training data for those genres and because Funk and Hip Hop tend to have a more subtle swing than Jazz and Rock. Furthermore, the second model was trained for 200 epochs instead of just 50 in attempt to increase its accuracy. Results of both models are reported in this section with results of the first iteration labeled T2G-1 and the second iteration T2G-2. Due to time constraints, the listening test was only conducted with outputs from T2G-1.

A. Objective Evaluation

Table I shows the reconstruction metrics of offsets O and velocities V for both models. In general, T2G-2 slightly outperforms T2G-1. Both models achieved low masked Huber losses, with T2G-1 at 0.002 and T2G-2 at around 0.001, indicating accurate reconstruction of both expressive dimensions. For timing offsets, the models have an average timing error of about 4 to 5 ms, suggesting that both models are able to reproduce the microtiming characteristics of the target drum performances with relatively small deviations. Velocity reconstruction showed similar performance, T2G-1 has a MAE of approximately 4.87 MIDI units on the full 128 unit MIDI velocity range while T2G-2 has a MAE of 3.80 MIDI units.

Model	Dynamic	Mean Velocity	Threshold	Accuracy
T2G-1	Soft	0.42232	< 0.48323	0.87283
	Loud	0.67703	> 0.62370	0.75714
	Medium	0.52776	otherwise	0.73170
T2G-2	Soft	0.42308	< 0.48323	0.87824
	Loud	0.68723	> 0.62370	0.78751
	Medium	0.53589	otherwise	0.76593

TABLE II

MEAN VELOCITY AND ACCURACY OF BOTH MODELS' OUTPUTS WHEN GIVEN PROMPTS THAT INDICATE SOFT, MEDIUM OR LOUD DYNAMICS, AS WELL AS THE DEFINED MEAN VELOCITY THRESHOLDS USED WHEN LABELING THE TRAINING DATA.

Model	Feel	MSR	STDO	Threshold	Accuracy
T2G-1	Swung	1.4232	0.2056	$MSR \geq 1.5$	0.4346
	Loose	1.0437	0.1186	$STDO \geq 0.1$	0.6641
	Tight	1.0325	0.0543	$STDO < 0.1$	0.9855
T2G-2	Swung	1.4310	0.2071	$MSR \geq 1.5$ or 1.2	0.5531
	Loose	1.0400	0.1305	$STDO \geq 0.1$	0.7615
	Tight	1.0319	0.0596	$STDO < 0.1$	0.9575

TABLE III

MEAN FEATURES AND ACCURACY OF BOTH MODELS' OUTPUTS WHEN GIVEN PROMPTS THAT INDICATE LOOSE, SWUNG, OR TIGHT FEELS, AS WELL AS THE DEFINED THRESHOLDS FOR EACH FEEL WHEN LABELING THE TRAINING DATA. MSR REFERS TO MEAN SWING RATIO AND STDO REFERS TO STANDARD DEVIATION OF OFFSETS. FOR T2G-2, SWING IS DEFINED AS ≥ 1.2 MEAN SWING RATIO FOR FUNK AND HIP HOP, ≥ 1.5 FOR ROCK AND JAZZ.

The Pearson correlation coefficients further demonstrate strong agreement between the predicted and test set performance parameters. Both models achieved a correlation of around 0.93 to 0.97 for offsets and velocities, indicating that the model also closely reproduces the relative variation in timing and dynamics across drum hits.

The results of the dynamic control tests are shown in Table II. It shows that for both models, the mean velocities of the soft, medium and loud outputs stay well within the boundaries we defined for dynamics when labeling the training data. However, when looking at the actual accuracy, which shows the percentage of outputs that have a mean velocity satisfying the requested dynamic, the results show that soft has the highest accuracy, with about 87% for both models, followed by loud and medium at around 73 to 79%. In general, both models were able to correctly predict outputs with the correct velocity characteristics, with T2G-2 performing slightly better.

The results of the feel control tests are shown in Tables III and VII. While both models have similar results for the mean features of their outputs, T2G-2 has a substantial increase in feel accuracy for loose and swing. Table III shows that the mean features of both models' outputs tend to correspond with the requested feel. As expected, swung outputs have the highest standard deviation of offsets at around 0.2 and the highest mean swing ratio at around 1.4, which does not quite reach the defined 1.5 threshold, meaning this area of the models still require further refinement. On the other hand, loose and tight has mean standard deviation of offsets at around 0.1 and 0.05, which satisfies the defined requirements. In terms of accuracy, T2G-2 performs much better overall,

Metric	T2G-1	T2G-2
Mean Offset Diversity (ms)	10.61734	12.08361
Mean Velocity Diversity (MIDI units)	16.29351	19.33201
Mean Pairwise Timing Difference (ms)	12.14066	13.84631
Mean Pairwise Velocity Difference (MIDI units)	18.15462	21.47585

TABLE IV

MEAN DIVERSITIES AND MEAN PAIRWISE DIFFERENCES WHEN CHANGING ONLY THE LATENT VARIABLE z .

Rating	Proposed vs.	Estimate [95% CI]	Adj. p
PM	Baseline	-3.43 [-8.34, 1.49]	.343
PM	Nearest Neighbour	0.06 [-4.88, 5.00]	.980
PM	Quantized	11.00 [6.07, 15.93]	$< .001$
HL	Baseline	-9.41 [-14.39, -4.43]	$< .001$
HL	Nearest Neighbour	0.36 [-4.65, 5.37]	1.000
HL	Quantized	-0.45 [-5.45, 4.55]	1.000

TABLE V

PROMPT-MATCHING AND HUMAN-LIKENESS RESULTS OF PROPOSED METHOD COMPARED TO BASELINE, NEAREST NEIGHBOUR, AND QUANTIZED METHODS. POSITIVE ESTIMATES FAVOUR THE PROPOSED METHOD. P-VALUES WERE HOLM-ADJUSTED WITHIN EACH OUTCOME.

with loose being 76% and swing being 55%, a 10% increase in both feels over T2G-1. This comes at the cost of a 3% decrease for tight but this difference is not significant as T2G-2's tight accuracy is still very high, at 95.75%. While T2G-2 can correctly predict the characteristics of tight and loose well, swing feel still needs improvement.

Finally, the results of the latent diversity test in Table IV show that across outputs with varying z , a hit on average differs by 10 to 12 milliseconds in terms of offset and 16 to 19 MIDI units in terms of velocity. Furthermore, two independently generated performances differed on average by 12 to 13 ms in timing and 18 to 21 MIDI velocity units in dynamics. This shows that for both models, different latent samples produce distinct timing and velocity realizations under the same conditioning, with T2G-2 being slightly more diverse.

B. Listening Test

Table V presents the overall method contrasts from the mixed-effects model. For prompt matching, the proposed method received significantly higher ratings than the quantized method (adjusted $p < 0.001$), but did not differ significantly from the baseline or nearest neighbour methods. For human-likeness, the proposed method was rated significantly lower than the baseline (adjusted $p < 0.001$), with no significant

Rating	Interaction with	LR χ^2	df	Adj. p
PM	Genre	13.20	9	.154
PM	Feel	16.53	6	.022
PM	Dynamics	25.37	6	$< .001$
HL	Genre	12.37	9	.193
HL	Feel	12.07	6	.121
HL	Dynamics	18.22	6	.017

TABLE VI

INTERACTIONS BETWEEN METHOD AND STIMULUS GENRE, FEEL AND DYNAMICS CHARACTERISTICS.

differences from the nearest-neighbour or quantized methods. The interaction analysis in Table VI shows that differences in prompt-matching ratings between methods varied significantly by feel (adjusted $p < 0.022$) and dynamics (adjusted $p < 0.001$), but not genre. For human-likeness, method differences varied significantly by dynamics (adjusted $p < 0.017$) but not genre and feel.

V. DISCUSSION

Overall, Text2Groove is good at reconstructing human performance parameters but the lower conditioning accuracies suggest that it has difficulties obeying semantic controls reliably. With that being said, dynamics control still performed reasonably well. This is likely due to the fact that different dynamics can be achieved by simply scaling the velocities up or down. It is much easier to achieve than shifting offsets while considering the complex relationships between timings of multiple notes. On the other hand, feel control struggled, particularly with predicting swung outputs. T2G-2 only had a swung accuracy of 55.31% but a 95.75% tight accuracy. This significant difference is likely due to the fact that tight performances are much easier to express statistically, whereas swing requires systematic relationships between particular pairs of on-beat and off-beat events. Furthermore, 8th note swing and 16th note swing are both simply classified as swing in the training data, despite large differences in characteristics. This could have contributed to the low accuracy of the model and should be addressed in future work by treating them as two separate categories.

We can further pinpoint the models' weaknesses by looking at Table VII, which shows the mean swing ratio, mean std offset and feel accuracy of both models' outputs categorized by genre and feel. The feel accuracy section shows the improvements in swung Jazz, swung Rock, and loose feels overall when comparing the two models, with swung Rock being the most significant, from 67.60% to 94.87%. These improvements are likely due to the correct labeling of training data. However, even after loosening the swing ratio threshold for Funk and Hip Hop and the correct labeling of data, swung Funk and Hip Hop still had poor accuracy, at 31.28% and 11.05% respectively. While the low swung Funk and Hip Hop accuracy for T2G-1 can be attributed to incorrect labeling and lack of Hip Hop training data, the same cannot be said for T2G-2. For the second model, swung Funk had the most samples out of all swung genres with 8343 (see Table VIII for the full distribution for both models), yet had much worse accuracy than swung Rock, which only had 3301 samples but an accuracy of 94.87%. One possible explanation is that since the swing ratio threshold was lowered to 1.2, there was a large range of degrees of swing in the training data. Patterns with only subtle swing could therefore be grouped together with patterns exhibiting much stronger swing, resulting in a less consistent relationship between the swung conditioning label and the timing characteristics the model was expected to learn. In contrast, swung Rock retained the higher threshold of 1.5, meaning that its training examples may have contained more

clearly and consistently swung timing patterns. The lower threshold may have increased the number of available swung Funk and Hip Hop examples, but at the same time made the category less distinct from tight or loose performances.

Another limitation regarding the current definitions of the different feels is that standard deviation of offsets captures the overall magnitude of timing variation but does not describe where deviations occur or whether they follow musically meaningful patterns. For example, two patterns can have the same standard deviations of offsets but one could be deliberately laid-back, while the other contains unintended human errors. This makes it difficult for the model to learn the characteristics of what loose means.

Perhaps the biggest limitation of this method is the synthesized training data. While this gives us counterfactual training pairs with the same score but different requested groove, the transferred timing and velocity is not guaranteed to be musically appropriate for the target score, especially because expressive performance and hit pattern are not truly independent (Danielsen, Waadeland, et al. 2015; Câmara et al. 2020). This is supported by results of the listening test that showed both the proposed (95% CI [-14.39, -4.43], adjusted $p < 0.001$) and nearest neighbour (95% CI [-14.75, -4.79], adjusted $p < 0.001$) methods being rated significantly lower than baseline in terms of human-likeness. This suggests the quality and musical compatibility of the synthesized training data represents an important bottleneck in the Text2Groove pipeline.

Using the results of the listening tests, we can identify condition-specific weaknesses of Text2Groove. Table IX shows that the proposed method received substantially lower human-likeness ratings than baseline for soft performances (adjusted $P < 0.001$). This is likely because mean velocity was used to label dynamic level, so patterns can still contain a few loud hits that change the perceived dynamic level. Similarly, results of the loud condition also show a similar trend likely due to the same mean velocity limitation, but cannot be confirmed without more participants (adjusted $p = 0.220$). Lower human-likeness ratings were also observed for swung and loose performances, which can be attributed to labeling and synthesized data limitations described above. However, these results also require more participants as the overall method and feel interaction was not significant (adjusted $p = 0.121$).

VI. FUTURE WORK

Future work should primarily address the limitations observed in semantic control and perceptual quality of the outputs. Results suggest that representing swing as a discrete label defined by arbitrary thresholds may be insufficient. Rather than classifying performances as swung or not swung, future versions could condition the model on mean swing ratio to increase its accuracy. Another benefit of this approach is that users would be able to explicitly control the degree of swing the output should have. Furthermore, patterns with 8th note swing and 16th swing should be classified as different

Model	Genre	Mean Swing Ratio			Mean Std Offset			Feel Accuracy		
		Loose	Swung	Tight	Loose	Swung	Tight	Loose	Swung	Tight
T2G-1	Funk	1.05513	1.28803	1.03560	0.10900	0.18463	0.05573	59.78	31.62	98.00
	Hip Hop	1.02912	1.14896	1.04483	0.13057	0.13210	0.06223	75.40	5.35	96.91
	Jazz	1.07967	1.55415	1.03944	0.12570	0.22692	0.05908	75.56	57.91	97.71
	Rock	1.03191	1.61770	1.03297	0.11730	0.25770	0.04470	65.42	67.60	98.87
T2G-2	Funk	1.03119	1.16675	1.02647	0.11910	0.16021	0.06129	71.07	31.28	92.68
	Hip Hop	1.00831	1.06025	1.02904	0.14533	0.14114	0.06529	83.97	11.05	94.58
	Jazz	1.07666	1.60560	1.03599	0.13476	0.23644	0.05983	77.16	72.07	96.96
	Rock	1.03560	1.79436	1.03556	0.12791	0.27335	0.05362	74.95	94.87	98.58

TABLE VII

MEAN SWING RATIO, MEAN STANDARD DEVIATION OF OFFSETS AND FEEL ACCURACY OF BOTH MODELS' OUTPUTS CATEGORIZED BY GENRE AND FEEL.

Genre	T2G-1			T2G-2		
	Loose	Swung	Tight	Loose	Swung	Tight
Funk	14225	7883	14075	14182	8343	14075
Hip Hop	12160	942	11048	12068	2290	11034
Jazz	15444	11034	13809	15449	4779	13814
Rock	16127	12291	15658	16148	3301	15683

TABLE VIII

GENRE AND FEEL DISTRIBUTION COUNTS OF THE TRAINING DATA USED FOR T2G-1 AND T2G-2.

Rating	Condition	Prop.	Base	Difference [95% CI]	Adj. <i>p</i>
HL	Soft	43.9	59.0	-15.1 [-23.0, -7.3]	< .001
HL	Swung	36.4	50.5	-14.1 [-22.4, -5.8]	.002*
HL	Loose	43.8	53.9	-10.1 [-18.3, -2.0]	.046*
PM	Loud	44.5	52.5	-8.0 [-16.8, 0.8]	.220 ^a

TABLE IX

CONDITIONS IDENTIFIED FOR FURTHER IMPROVEMENT WHEN COMPARING PROPOSED AND BASELINE METHODS.

*REQUIRES CONFIRMATION WITH ADDITIONAL PARTICIPANTS.

conditions as the locations of the on beats and off beats are different. Future work should also define the characteristics of loose more clearly instead of using standard deviation of offsets. One possible improvement is to use a swing ratio of around 1.17 to define loose. This stems from an interview with the creator of electronic drum machine swing, Roger Linn, who stated that “a swing setting of 54% [corresponding to a swing ratio of 1.17] will loosen up the feel without it sounding like swing” (Scarth and Linn 2013). To tackle the drawbacks of synthesized data and improve perceptual quality of outputs, ideal future work would be to collect new data from human drummers playing the same scores in several different styles. However, this is not a very scalable solution and perhaps future work could instead focus on better methods of generating new data.

VII. ACKNOWLEDGMENTS

I would like to thank my supervisor, Georgios Velissaridis, for his guidance, feedback, and support throughout this research project. I am also grateful to Hidetaka Imamura for his contributions to the development of the online listening test and constructive feedback throughout the project.

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APPENDIX A
ETHICS APPROVAL DOCUMENTS



Participant Information Sheet

Study title

Rate how well drum grooves match a description

Version number and date

Version 1.0: 01/08/2026

Researcher's name

Kwok Tung Adrian Lam (Student Investigator)

Georgios Velissaridis (Supervisor)

Queen Mary Research Ethics Committee reference number:

QME26.070

Invitation paragraph

You are being invited to participate in a research study. Before you decide whether you wish to participate in this study, it is important for you to understand why the research is being done and what it will involve. Please take time to read the following information carefully and discuss it with others if you wish. Ask us questions if there is anything that is not clear or if you would like more information.

What is the purpose of the study and what would taking part involve?

This study aims to evaluate how well listeners perceive the generated output from the proposed model as matching the given semantic description with regards to drum microtimings, as well as how human-like the generated output sounds. You will listen to four versions of a drum audio track, the original quantized version, then three audio tracks modified using different methods including the proposed method, and rate how well each audio track matches the given semantic description. Then you will rate how likely the generated outputs are to be played by human drummers. Some demographic information will be collected at the end of the study. The process takes 15-25 minutes, is completed entirely online, and requires no follow-up sessions. Your responses will help us evaluate the generated output from our proposed drum groove editing model.

Why am I being invited?

You have been invited to participate because we are seeking musicians who understand the terms that describe groove (swing, tight, laidback etc.)

Do I have to take part?

This participant information sheet is here to help you decide whether you would like to take part in our study. Participation is entirely voluntary, and it is completely up to you whether or not to take part. If you choose to participate, you are free to withdraw at any time during the data collection phase without providing a reason and without any penalties. However, please note that once the data collection phase is complete, withdrawal will no longer be possible, since data will be stored anonymously. You can also skip any optional question you do not wish to answer without explanation.

What are the possible benefits of taking part?

By participating in this research, you will contribute to evaluating a text guided drum groove editing system, which allows beginner music producers to edit their drum grooves through text and speed up the process for expert producers.

What are the possible disadvantages and risks of taking part?

Taking part in this study involves minimal risks. You may experience mild fatigue from listening to the audio tracks, or completing the questionnaire, but you are free to pause or withdraw at any time. Your data will be anonymized and stored securely to protect your confidentiality. If you have any concerns, you can contact the research team.

Expenses and payments

This is an online survey. Participants are not expected to have any expense and they will not be reimbursed for their time.

What information about me will you be collecting?

We will collect basic demographic information (e.g., age, musical experience), and your preferences between tags. All responses will remain anonymous and will not include any personal identifying information.

How will my data be stored and who will have access to it?

Your data will be stored in a fully anonymized format on secure servers, accessible to the research team and to other researchers that request it. No personal identifiers will be collected, ensuring that your responses cannot be linked back to you. Data will be encrypted and managed following Queen Mary's Data Protection Policy and Information/Data Governance Policy (DG14). Anonymized data may be retained for future research purposes in accordance with ethical guidelines.

When and how will my data be destroyed?

Your data will be securely stored in anonymized form for a period of 5 years in accordance with the Queen Mary Records Retention Schedule. After this period, all data will be permanently deleted using secure data destruction methods outlined in the university's Information/Data Governance Policy (DG16). If anonymized data is used for future research, it will be retained indefinitely but will remain untraceable to individual participants.

How will my data be used and shared?

Your data will be reported in a master's dissertation and may be part of publications in academic conferences, in fully anonymised format. The data will be stored in a database accessible to the research team, and may be shared in anonymised form with other researchers to support future research.

Under what legal basis are you collecting this information?

- Queen Mary University of London **processes personal data** for research purposes in accordance with the lawful basis of 'public task'.
- Please read [Queen Mary's privacy notice for research participants](#) containing **important information about your personal data and your rights** in this respect.
- **If you have any questions** relating to data protection, please contact Queen Mary's Data Protection Officer, Queens' Building, Mile End Road, London, E1 4NS or data-protection@qmul.ac.uk or 020 7882 7596.

What will happen if I want to withdraw from this study?

You can withdraw from this study at any time during the data collection phase without

providing a reason. If you choose to withdraw, any data you have provided will be confidentially destroyed and not used in the research. However, after submission, it will no longer be possible to withdraw your data as it will have been anonymised.

What should I do if I have concerns about this study?

- **If you have any concerns** about the manner in which the study was conducted, in the first instance, please **contact the researcher(s)** responsible for the study Georgios Velissaridis, g.velissaridis@qmul.ac.uk
- **If you have a complaint** which you feel you cannot discuss with the researchers then you should **contact the Research Ethics Facilitators** by e-mail: research-ethics@qmul.ac.uk.
- When contacting the Research Ethics Facilitators, **please provide details** of the study title, description the study and QM REC reference number (where possible), the researcher(s) involved, and details of the complaint you wish to make.

Who can I contact if I have any questions about this study?

Kwok Tung Adrian Lam,
a.k.lam@se25.qmul.ac.uk

Georgios Velissaridis, g.velissaridis@qmul.ac.uk



Consent Form

Title of Research Study:

Rate how well drum grooves match a description

Principal Investigator:

Kwok Tung Adrian Lam

Georgios Velissaridis

Queen Mary Research Ethics Committee Ref:

QME26.070

Thank you for your interest in this research.

Should you wish to participate in the study, **please consider the following statements.**

Before signing the consent form, you should **initial all or any of the statements that you agree with.**

Your signature **confirms that you are willing to participate in this research**, however you are reminded that **you are free to withdraw your participation** at any time.

Statement	Please initial box
1. I confirm that I have read the Participant Information Sheet dated 01/08/2026 version 1.0 for the above study; or it has been read to me. I have had the opportunity to consider the information, ask questions and have had these answered satisfactorily.	
2. I understand that my participation is voluntary and that I am free to stop taking part in the study at any time during participation without giving any reason and without my rights being affected.	
3. I understand that my data will be accessed by the research team.	
4. I understand that my data will be securely stored in university servers and in accordance with the data protection guidelines of the Queen Mary University of London 5 years in fully anonymised form.	
5. I understand that I can access the information I have provided and request destruction of that information during the completion of the experiment. I understand that after submission, I will not be able to request withdrawal of the information I have provided.	
6. I understand that the information collected about me will be used to support other research in the future, and it may be shared in anonymised form with other researchers.	
7. I agree to take part in the above study.	

Participants should read [Queen Mary's privacy notice](#) for research participants which contains **important information about your personal data and your rights** in this respect.

If you have any **questions relating to data protection**, please contact Data Protection Officer, Queens' Building, Mile End Road, London, E1 4NS or data-protection@qmul.ac.uk or 020 7882 7596.

_____	_____	_____
Participant name	Date	Signature
_____	_____	_____
Name of person taking consent	Date	Signature

I, Kwok Tung Adrian Lam confirm that I have carefully explained the nature, demands and any foreseeable risks (where applicable) of the proposed research to the participant and provided a copy of this form.

**Principal Investigator (or Supervisor
for student projects)**

Georgios Velissaridis,

g.velissaridis@qmul.ac.uk

Student Investigator (if applicable)

Kwok Tung Adrian Lam,

a.k.lam@se25.qmul.ac.uk

**School of Electronic Engineering
& Computer Science**

Peter Landin Building, 10 Godward Square
London E1 4FZ

Devolved School Research Ethics Committee

Email: eeecs-ethics@qmul.ac.uk

22 June 2026

**Mr Georgios Velissaridis
School of Electronic Engineering and Computer Sciences
Queen Mary University of London**

Cc: Kwok Tung Adrian Lam

Dear Georgios Velissaridis

RE: DSEEC26.070 - Evaluation of Text2Groove, a text guided drum groove editing system.

Having received a Research Ethics Application for the study detailed above, the proposed work has been deemed low-risk and will not require the scrutiny of the full Queen Mary Research Ethics Committee. It has been reviewed by the Electronic Engineering and Computer Science Devolved School Research Ethics Committee (EECS DSREC) and I am pleased to inform you that, based on the information provided, this application has now been **approved**.

- **Date of Approval: 22 June 2026**
- **Approval End Date: 21 June 2029**

Extensions and Amendments:

If you propose to make an [amendment to the research after approval](#) or extend the study beyond the date of the approval expiry date given above, then the EECS DSREC should be contacted (eeecs-ethics@qmul.ac.uk).

Adverse Events:

In case of any unexpected or adverse events or deviations from the proposed protocol as per the original application, please contact the EECS DSREC at eeecs-ethics@qmul.ac.uk within 5 days of the event.

Health and safety:

Researchers should exercise extra care and precautions when conducting research at this time. There are ethical issues that are of heightened importance to consider when conducting research with human participants.

Researchers are responsible for checking the local and national governmental advice and for being aware of any necessary restrictions that may impact their study. In particular, research that involves overseas travel or face-to-face research must regularly check for updates and follow the latest guidance on Health and Safety.

We wish you the best for your research.

Yours faithfully,

Alvaro Bort

Alvaro Bort

On behalf of the Electronic Engineering and Computer Science DSREC

APPENDIX B
GENERATIVE AI DECLARATION

ECS750P/ECS753P/ECS754P– MSc Project Module
Generative AI Tools – Student Accountability Statement

The completed and signed statement must be included in the appendix of the final dissertation paper.

Student Name: Kwok Tung Adrian Lam

Project Title: Text2Groove: Text-Guided Drum MIDI Humanization

Supervisor Name: Georgios Velissaridis

1. Have you used Generative AI tools for your project?

YES

2. Please provide details of all the AI tools that you used in your project work.

AI Tool	Version	Publisher	Access URL
ChatGPT	5.6	OpenAI	https://chat.openai.com/

3. If you have used Artificial Intelligence (AI) tools in any part of your project please include in table format how, where, and why these tools were used. This should include references to the relevant parts of your project work — for example:

- Specific chapters or sections of the report (e.g. literature review, methodology, requirements)
- Software development (e.g. AI-assisted code generation, debugging, documentation)
- Interview or survey materials (e.g. AI-generated or refined questions/prompts)
- Datasets (e.g. AI-assisted data cleaning, labelling, or synthesis)
- The showcase video (e.g. AI-assisted scripting, narration, or editing support)

Be specific and transparent about your use of AI tools, including both direct outputs (e.g. text, code, images) and indirect uses (e.g. idea generation, editing assistance). The goal is to ensure academic integrity and provide a clear record of how AI contributed to your work.

Project Section	Description	Justification
Dissertation Paper	Grammar help	To identify and fix grammar mistakes since English is not my first language, and to rephrase sentences more coherently.
Listening test	Generate code for custom web interface and Reaper script to batch render audio files	The design of the study required a custom web interface and a way to render thousands of audio stimuli. We were already behind schedule so I used ChatGPT to help generate

		the code for it. This was discussed with my supervisor.
Statistical analysis	Assist with writing code for mixed effects model analysis	I have not done mixed effects model analysis before so I asked ChatGPT for help.
Project Code	General debugging and suggestions on how to improve	Got stuck a few times with the Python scripts and training code so I asked ChatGPT what the issue could be and how to improve it

3. Generative AI Student Declaration

- I have declared all uses of AI tools (e.g., coding assistants, text generators, data analysis tools) in this project.
- I have reviewed and verified all AI-assisted content to ensure accuracy and suitability.
- I take full responsibility for the originality and integrity of my project submission.
- I have referenced all sources, including those suggested or produced with AI support.
- I have not used AI in ways that undermine academic honesty (e.g., plagiarism, unverified code submission).
- My use of AI was supplementary and did not replace my own independent learning and problem-solving.

Signed: _____Adrian_____ Date: _____Aug 18 2026_____

Guidance notes for students

- Students are strongly encouraged to review the following resources
 - Student Guide to Gen AI <https://www.qmul.ac.uk/library/academic-skills/student-guide-to-generative-ai/>
 - Academic Integrity and Misconduct Policy <https://www.qmul.ac.uk/governance-and-legal-services/media/dgls-media/policy/current-policies/Academic-Integrity-&-Misconduct-Policy.pdf>
- The use of essay mills, homework help sites, plagiarism, collusion, falsification, impersonation, or actions that may give an unfair advantage is strictly prohibited in your final year project. Please refer to the QMUL Academic Integrity and Misconduct policy above.
- It is never acceptable to use GenAI tools to write your entire literature review and submit it as your own work; doing so is considered a form of plagiarism since GenAI outputs are derived from ideas of human authors without proper referencing.
- You may use GenAI tools for checking grammar and spelling, provided the tool is not rewriting a large portion of your work for your final year project.
- Using GenAI for idea generation or initial planning is generally appropriate but always consider the context and assessment requirements for your project. If in doubt, please talk to your supervisor.
- If you are ever unsure whether GenAI is permitted for any part of your project always ask your project supervisor for explicit clarification.
- Unless stated otherwise, GenAI should only be used to assist your learning at the initial stages of your project and your final submission of your project, and the code must still be substantially your own original work.

- Inappropriate use of GenAI will be viewed as plagiarism under the QMUL academic integrity and misconduct policy this includes submitting work where GenAI's contribution exceeds what is reasonably allowed in the final year project.
- Attempting to pass off GenAI-generated content as your own can result in disciplinary investigation under academic misconduct regulations; you may be asked to attend a viva to confirm authorship.
- Before using GenAI, make sure:
 - You are permitted (or required) to use it for your specific task in your final year project.
 - You fully understand the potential risks, limitations, and how GenAI operates.
 - Your work remains genuinely your own and reflects your understanding.
- Please note that GenAI is helpful for gathering background information, but outputs may be biased, inaccurate, outdated, or fabricated.
- Always consult and cite original, credible sources; GenAI should never be treated as an original source and should not be cited as an author or in the reference list.
- Critically assess all GenAI output for accuracy, double-check any facts, quotations, or citations it suggests especially if you use them in your literature review or other chapters.
- If GenAI has been used for drafting, planning, or structuring, you must clearly acknowledge and document its use in your submission. Therefore, you should keep drafts and records of your own process to show your independent scholarly work and reading notes
- GenAI use may not require acknowledgement for:
 - Familiarising yourself with a topic or research landscape.
 - Summarising information to prepare for more in-depth research.
 - Improving grammar or writing structure.
- Please note that output from GenAI is typically treated as having no author so you cannot cite Gen AI as an author in your report
- When in doubt, transparency and honesty about your methods and sources including GenAI are expected and protect your academic integrity